UPC ϕ Meson Analysis Note

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Contents

1 Introduction 2
2 Dataset and Trigger 2
2.1 Luminosity 4
2.2 Bad Runs 5
3 Event Selection 5
3.1 Vertex Selection 6
3.2 BBC Veto 6
3.3 Event Multiplicity Selection 7
4 Track and Pair Selection 7
4.1 Track Quality 8
4.2 Coherent and Incoherent Regions 10
4.3 Particle Identification 11
4.4 Selection Summary 16
5 Signal Extraction 16
5.1 Coherent Signal 16
5.2 Incoherent Signal 16
5.3 Yield Summary 16
6 Efficiencies 16
6.1 Vertex Selection Efficiency 16
6.2 BBC Veto Selection Efficiency 23

1 Introduction

In this analysis, we measure coherent and incoherent ϕ meson photoproduction in ultra-peripheral Au+Au collisions (UPCs) at STAR. The ϕ mesons are reconstructed through the charged-kaon decay channel, $\phi \to K^+K^-$. The primary goal of the analysis is to extract the differential production cross sections for the coherent and incoherent components.

2 Dataset and Trigger

This analysis uses the Run-19 Au+Au dataset collected at $\sqrt{s_{NN}} = 200$ GeV. Run-19 is the first Au+Au dataset collected after the full installation of the inner Time Projection Chamber (iTPC) upgrade. The improved inner-sector acceptance allows charged tracks to be reconstructed down to transverse momenta of approximately $p_T \sim 60$ MeV/c. This is particularly important for this analysis because the decay kaons are soft. The ϕ carries very little momentum, especially in coherent production, and in the ϕ rest frame each kaon has a momentum

$$p^* = \sqrt{\left(\frac{m_\phi}{2}\right)^2 - m_K^2} \approx 127 \text{ MeV}/c, \tag{1}$$

so in the laboratory frame the kaon transverse momenta are of order 100 MeV/c.

The dataset used in this analysis has the following configuration:

- Production library: `SL23e`;
- Data format: `MuDst`.

No dedicated UPC trigger was available for this dataset. Instead, events are selected from the Zero Degree Calorimeter (ZDC) minimum-bias (zdcmb) trigger sample.

In UPCs of heavy nuclei, virtual photon exchanges can electromagnetically excite one or both nuclei, most commonly through the Giant Dipole Resonance (GDR). The excited nucleus subsequently de-excites through the emission of one or more forward neutrons, which remain close to beam rapidity and are detected by the ZDCs. The zdcmb trigger therefore provides a baseline sample of UPC events accompanied by mutual electromagnetic dissociation.

The trigger configuration is summarized below:

- Offline trigger ID: `700001`;
- Trigger requirements: ZDC-East, ZDC-West, and ZDC TAC coincidence, with no minimum Time-of-Flight (TOF) requirement;
- Number of recorded events: 138.38 million.

The absence of a TOF requirement is essential for this analysis: as discussed above, the decay kaons are largely too soft to reach the TOF, so a trigger requiring TOF hits would remove most of the coherent signal.

The analyzed dataset was recorded with ZDC coincidence rates ranging from 1.5 kHz to 5.0 kHz, as shown in Figure 1. This corresponds to a clean, low-luminosity running condition. At this collision rate, the mean time separation between collisions is 200–700 µs, which is much longer than the TPC drift time of ~ 40 µs.

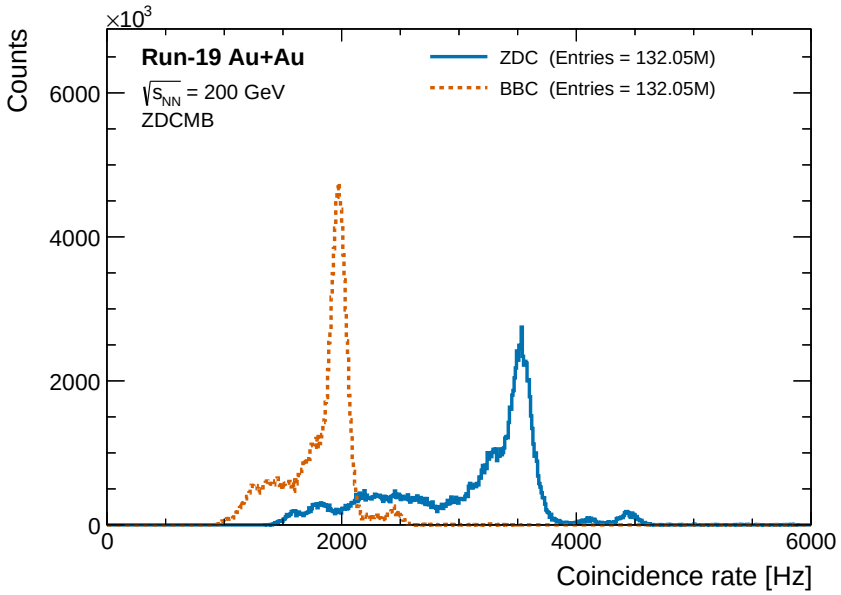


Figure 1: ZDC and BBC coincidence rates for the Run-19 Au+Au zdcmb trigger.

In the Run-19 trigger configuration, there is no cut on the ZDC ADC values, so no ZDC selection efficiency is applied in this analysis. The unattenuated ZDC ADC values that saturate the 12-bit ADC range are placed in the overflow bin, which appears as a spike at the right edge of the ZDC ADC distribution in Figure 2.

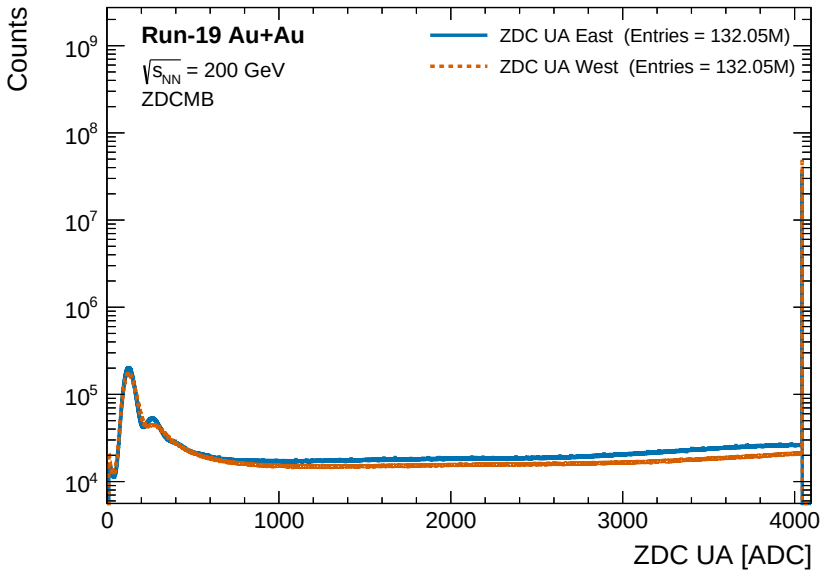


Figure 2: ZDC unattenuated ADC values for the Run-19 Au+Au zdcmb trigger.

2.1 Luminosity

The integrated luminosity for the zdcmb trigger is obtained from the STAR luminosity database for Run-19, where it is determined from the ZDC coincidence rate and the ZDC coincidence cross section. The corresponding net cross section, extracted as the slope of the live ZDC coincidences in each run, $\langle R_{\mathrm{ZDC}} \rangle \times t_{\mathrm{run}} \times$ livetime, versus the database luminosity, is 12.2 b (Figure 3); here t_{run} is the run duration. The number of recorded events versus the integrated luminosity for each run is shown in Figure 4 and exhibits a strong linear correlation, indicating that the luminosity determination is consistent for all analyzed runs. The integrated luminosity for the zdcmb trigger before bad-run removal is $L_{\mathrm{int}} = 13.2666\,\mu\mathrm{b}^{-1}$, with a prescale factor of 1.

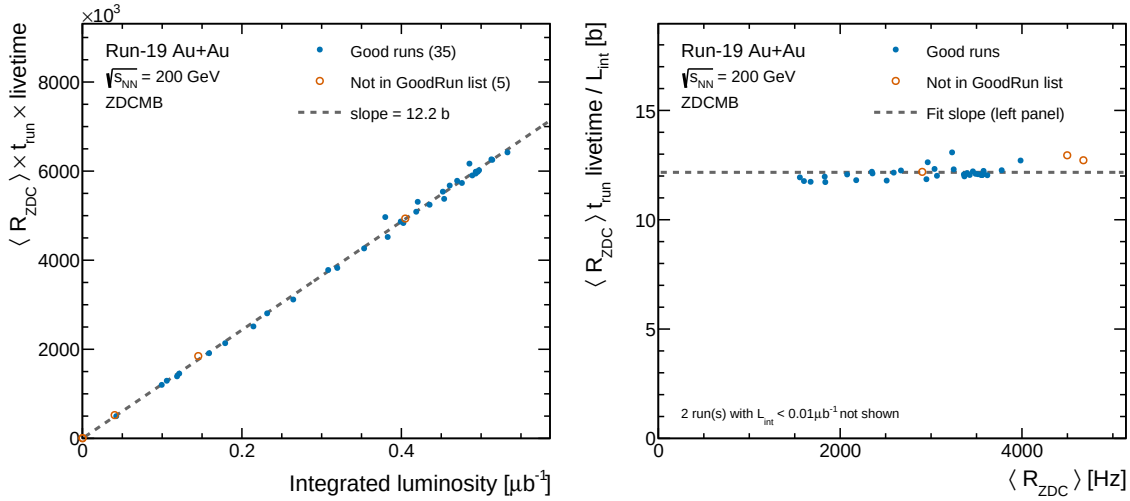


Figure 3: Run-by-run integrated luminosity compared to the ZDC coincidence rate for the Run-19 Au+Au zdcmb trigger. Left: live ZDC coincidences per run versus integrated luminosity, with a proportional fit. Right: their ratio, the implied ZDC coincidence cross section, versus the mean ZDC coincidence rate.

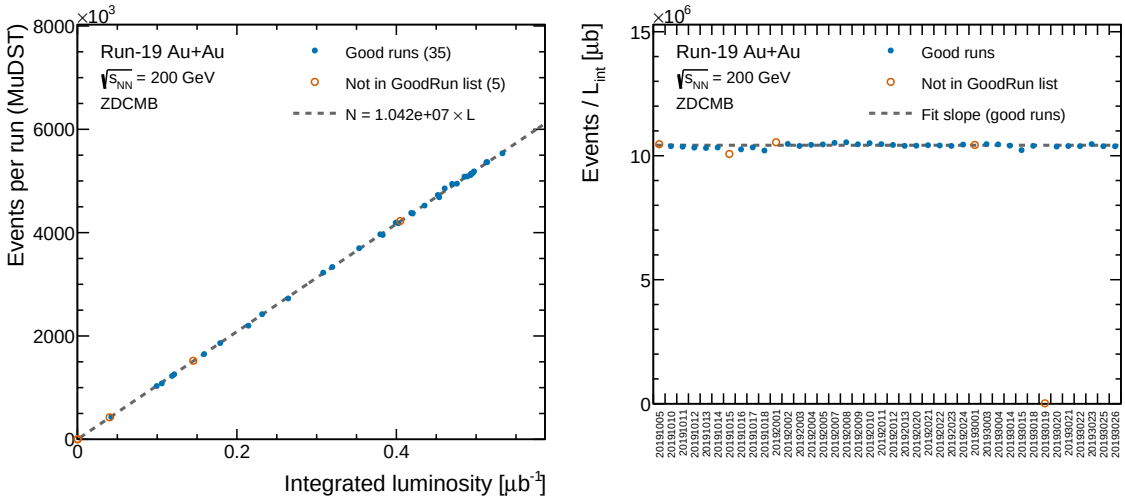


Figure 4: Run-by-run integrated luminosity compared to the number of events recorded for the Run-19 Au+Au zdcmb trigger. Left: events per run versus integrated luminosity, with a proportional fit. Right: events per unit luminosity for each run.

2.2 Bad Runs

Prior to applying event-level selections, runs with known detector-quality issues are removed from the analysis. The standard STAR BES-II Run QA bad-run list for Run-19 Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV is used to exclude periods with degraded TPC performance. Table 1 lists the run-by-run integrated luminosities for the bad runs.

Table 1: Integrated luminosity associated with the BES-II QA bad runs removed from the Run-19 Au+Au dataset.

Run number	Integrated luminosity [μb^{-1}]
20191005	0.145248
20191015	0.000152
20192001	0.040369
20193001	0.404860
20193019	0.000067
Total	0.590695

The nominal integrated trigger luminosity after bad-run removal for this analysis is

$$L_{\text{nominal}} = 12.6759\,\mu b^{-1} \tag{2}$$

3 Event Selection

UPCs are characterized by large impact parameters, which strongly suppress direct hadronic interactions between the two nuclei. In photon-induced vector-meson production, the exchanged photon is colorless and the interaction with the target proceeds through a

color-singlet exchange. Consequently, no color connection is established between the two beam nuclei, suppressing particle production between the centrally produced vector meson and the outgoing nuclear remnants. This gives rise to large rapidity gaps on either side of the produced ϕ meson.

We exploit this topology to suppress hadronic Au+Au backgrounds. The rapidity-gap requirement is implemented through a veto on forward charged-particle activity in the Beam-Beam Counters (BBCs), while the exclusive nature of the final state is further selected using low central-event multiplicity requirements.

3.1 Vertex Selection

The reconstructed primary vertex is used to identify the collision location and to ensure that selected events originate within the nominal STAR interaction region. Since the geometrical acceptance and tracking performance of the TPC depend on the vertex position, particularly along the beam direction, events reconstructed far from the detector center may exhibit reduced or non-uniform tracking efficiency. Vertex-quality requirements are therefore applied to define a well-controlled collision region and to suppress poorly reconstructed or non-collision vertices before the remaining event-level selections are imposed.

For this analysis, we require the reconstructed event vertex to satisfy

$$|V_z| < 100\,\text{cm}. \tag{3}$$

The integrated luminosity is determined using the ZDC-based luminosity measurement, whose longitudinal acceptance is not restricted in the same way as the TPC tracking acceptance. Because the effective fiducial region of the TPC is smaller than the longitudinal interaction region sampled by the ZDC luminosity measurement, the V_z requirement introduces an additional fiducial acceptance on the analyzed event sample. The corresponding efficiency associated with this restricted vertex region, including the V_z-dependent vertex reconstruction and tracking efficiency, is therefore evaluated separately using embedding and incorporated into the analysis correction.

3.2 BBC Veto

The BBCs are forward scintillator detectors located on the east and west sides of STAR, approximately 3.7 m from the interaction point. The small (inner) BBC tiles are available in this dataset and cover the pseudorapidity range $3.3 < |\eta| < 5.0$. The BBCs are therefore sensitive to charged-particle production in the forward region. In this analysis, BBC activity is used as a veto to select the large rapidity gaps characteristic of UPC events and to suppress backgrounds from hadronic Au+Au collisions.

In past STAR UPC triggers, the BBC-veto online trigger was utilized for rapidity gap enforcement with the small BBC tiles. For each of the 18 small-tile PMTs on a given side, a BBC hit was defined by:

$$\text{ADC}_{\text{PMT}} > 20 \tag{4}$$

together with the timing requirement

$$100 \leq \text{TDC}_{\text{PMT}} \leq 2300. \tag{5}$$

The qualifying PMT ADC signals (those with a BBC hit) were summed separately in the East and West,

$$S_{\text{East}} \equiv \sum_{e\in \text{ hits}} \text{ADC}_e, \quad S_{\text{West}} \equiv \sum_{w\in \text{ hits}} \text{ADC}_w, \tag{6}$$

and the UPC-Main online veto threshold was 50 ADC channels on each side,

$$S_{\text{East}} \leq 50 \quad \wedge \quad S_{\text{West}} \leq 50. \tag{7}$$

The zdcmb trigger used in this analysis does not require this veto, but the BBC conditions are still evaluated at Level 0 and recorded with every event. The online BBC veto decision is obtained directly from the Level-0 trigger information stored in `StTriggerData`. For the Run-19 UPC zdcmb configuration, the East and West BBC conditions correspond to global DSM bits 17 and 18, respectively. Since `lastDSM(1)` contains global trigger bits 16–31, these correspond to bits 1 and 2 of the returned 16-bit word. The quantities are extracted as

$$\texttt{bbcEast} = (\texttt{lastDSM(1)} >> 1)\ \&\ 1, \qquad \texttt{bbcWest} = (\texttt{lastDSM(1)} >> 2)\ \&\ 1. \tag{8}$$

A value of one indicates that the corresponding BBC activity condition fired, i.e. that the small-tile BBC signal exceeded the online veto threshold on that side. An event therefore passes the BBC veto when neither the East nor the West bit is set. This analysis applies exactly this online requirement offline,

$$\texttt{bbcEast} = 0 \qquad \text{and} \qquad \texttt{bbcWest} = 0. \tag{9}$$

3.3 Event Multiplicity Selection

To select events consistent with the exclusive two-track final state of $\phi \to K^+K^-$, an event-multiplicity requirement is applied in addition to the BBC veto. The corresponding selections are:

- Each event must contain exactly two primary tracks.
- The two primary tracks must be associated with the same reconstructed primary vertex.

These requirements select events with a single two-track primary vertex and suppress events containing additional charged-particle activity, as expected from hadronic interactions or pileup.

4 Track and Pair Selection

After the event selection, each candidate event contains a single primary vertex with exactly two primary tracks. This section studies these tracks before any invariant-mass information is used. The aim is to characterize the overall track sample, namely its tracking quality, kinematics, and specific energy loss ($\mathrm{d}E/\mathrm{d}x$) in the TPC, and to define the track and pair requirements applied before signal extraction. Since the final state is reconstructed from only two tracks in an otherwise empty detector, the track sample is already relatively clean. We therefore apply a minimal set of track-quality requirements, chosen to retain as many good tracks as possible while rejecting poorly reconstructed ones.

Particle identification is based on the TPC $\mathrm{d}E/\mathrm{d}x$. For a track i and a particle hypothesis $X \in \{e, \pi, K, p\}$, the normalized deviation of the measured energy loss from the expected one is

$$n\sigma_X^{(i)} = \frac{1}{\sigma_{\mathrm{d}E/\,\mathrm{d}x}} \ln \frac{(\mathrm{d}E/\,\mathrm{d}x)^{(i)}_{\mathrm{meas}}}{(\mathrm{d}E/\,\mathrm{d}x)^{(i)}_X}, \tag{10}$$

where $(\mathrm{d}E/\,\mathrm{d}x)^{(i)}_X$ is the Bichsel expectation for hypothesis X at the momentum of track i and $\sigma_{\mathrm{d}E/\,\mathrm{d}x}$ is the $\mathrm{d}E/\,\mathrm{d}x$ resolution. For the two tracks of a pair, the compatibility with the hypothesis that both tracks are of type X is quantified by

$$\chi^2_{XX} = \left(n\sigma_X^{(1)}\right)^2 + \left(n\sigma_X^{(2)}\right)^2, \tag{11}$$

which is evaluated for the kaon (χ^2_{KK}), pion ($\chi^2_{\pi\pi}$), electron (χ^2_{ee}), and proton (χ^2_{pp}) hypotheses.

All distributions shown in this section share the following preselection:

- ZDC minimum-bias trigger (700001);
- bad runs removed;
- exactly two primary tracks associated with the pair's vertex;
- exactly two primary tracks in the event;
- pair $\mathrm{d}E/\,\mathrm{d}x$ compatibility with the kaon–kaon hypothesis, $\chi^2_{KK} < 100$ (Equation 11);
- track $p_T < 1\,\mathrm{GeV}/c$;
- pair $p_T < 1\,\mathrm{GeV}/c$;
- pair invariant mass, computed with the kaon mass hypothesis, $M_{KK} < 1.2\,\mathrm{GeV}/c^2$;
- no TOF requirement.

The two tracks form an unlike-sign (ULS) pair, the signal sample, or a like-sign (LS) pair, which is kept as a background reference.

4.1 Track Quality

4.1.1 Primary-Vertex Reconstruction

Primary vertices in this production (SL23e) are reconstructed with the Minuit vertex finder (`StMinuitVertexFinder`), which fits the vertex position from the distances of closest approach of TPC global tracks. Only global tracks passing `StMinuitVertexFinder::accept()` enter the fit:

```
bool StMinuitVertexFinder::accept(StTrack* track) const
{
  return (track &&
          track->flag() >= 0 &&
          track->fitTraits().numberOfFitPoints() >= mMinNumberOfFitPointsOnTrack
&&
          !track->topologyMap().trackFtpc() &&
          finite(track->length()) &&
          track->geometry()->helix().valid());
}
```

The cut values come from the `Calibrations/tracker/PrimaryVertexCuts` database table (Table 2).

Table 2: Track and vertex requirements of the Minuit vertex finder for the Run-19 Au+Au production.

Parameter	Value	Requirement
`MinNumberOfFitPointsOnTrack`	20	global-track fit points $N_{\mathrm{fit}}^{\mathrm{glob}} \geq 20$
`RImpactMax`	6 cm	transverse distance of the global helix to the beamline $< 6\,\mathrm{cm}$
`DcaZMax`	6 cm	track z at closest approach within 6 cm of the vertex seed
`MinTrack`	2	at least two accepted tracks per vertex
`ZMin`, `ZMax`	-200, 205 cm	range of the vertex search along z
`MinTrackPt`	0	not applied
`MinFracOfPossFitPointsOnTrack`	0	not applied

Since each selected vertex has exactly two tracks, both must pass $N_{\mathrm{fit}}^{\mathrm{glob}} \geq 20$. The primary refit adds the vertex as a fit point, giving the sharp edge at $N_{\mathrm{fit}} = 21$ (Figure 5, Figure 6); the small tail below it consists of tracks that lost hits in this refit. We therefore adopt

$$N_{\mathrm{hits}} \geq 20 \tag{12}$$

as the nominal track-quality selection, with $N_{\mathrm{hits}} = N_{\mathrm{fit}} - 1$ the number of TPC fit points. Its efficiency is evaluated with embedding.

The vertex finder applies **no minimum p_T requirement**: `MinTrackPt` is zero and is not read by the Minuit finder. The only implicit low-p_T limit is that strongly curling tracks may not reach 20 fit points; this is included in the embedding efficiency.

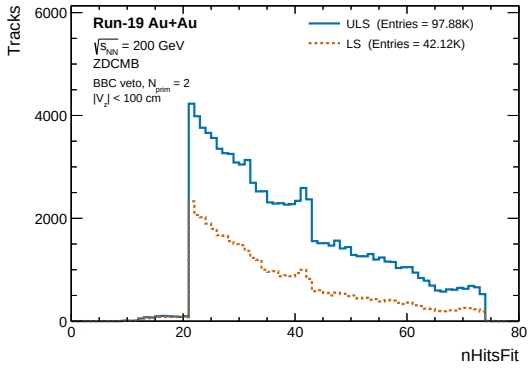


Figure 5: Number of fit points of primary tracks in unlike-sign (ULS) and like-sign (LS) pairs, after the BBC veto, $N_{\mathrm{prim}} = 2$, and $|V_z| < 100\,\mathrm{cm}$.

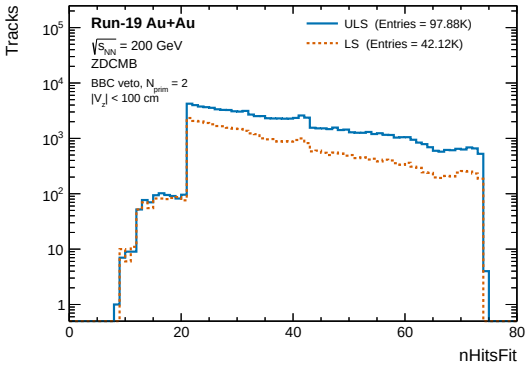


Figure 6: Same as Figure 5 on a logarithmic scale.

4.1.2 Inner TPC Upgrade and the Hit Requirement

Run 19 is the first run with the full inner TPC (iTPC), which increased the inner-sector pad rows from 13 to 40 and the maximum number of points per track from 45 to 72. The iTPC is fully operational in this sample: N_{fit} extends well beyond 45 (Figure 5), N_{poss} peaks at the full inner-sector and full-TPC lengths (Figure 7), and nearly all tracks have iTPC hits (Figure 8).

This makes $N_{\text{hits}} \geq 20$ a mild requirement here, although it is tighter than the 15 hits common in STAR UPC analyses. Many tracks, mostly at large $|\eta|$, leave through the endcap without reaching the outer sectors; with 40 inner rows they still collect 20 hits, which produces the edge at 20 in Figure 8, whereas with 13 inner rows they could not.

The fit-point threshold is not new: `MinNumberOfFitPointsOnTrack` is 20 also in 2012, 2014, and 2018. What changed is `MinTrack`, which was 5 in those years and is 2 here, so the Minuit finder can now reconstruct exclusive two-track events.

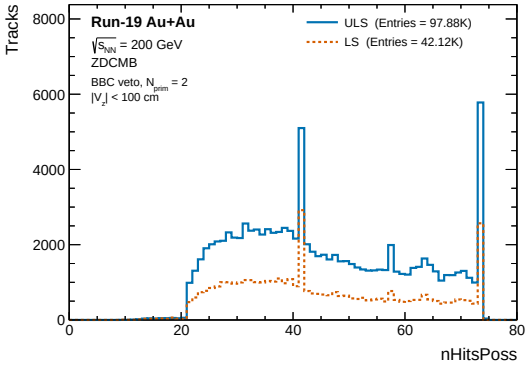


Figure 7: Number of possible points of primary tracks in ULS and LS pairs, same selection as Figure 5.

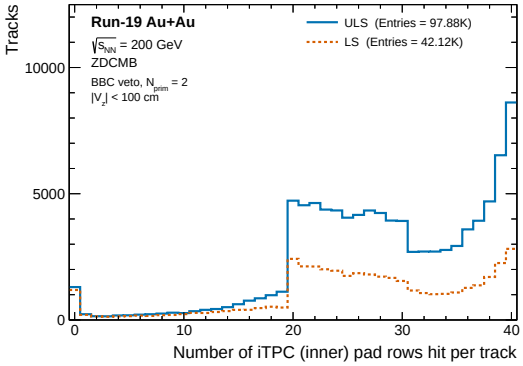


Figure 8: Number of iTPC pad rows hit per track, same selection as Figure 5.

4.2 Coherent and Incoherent Regions

In coherent production the photon couples to the nucleus as a whole, so the pair p_T is limited by the nuclear size, $p_T \lesssim \hbar c/R_{\text{Au}}$; incoherent production on individual nucleons extends to much larger p_T. We define the two regions by a cut on the pair p_T alone:

$$\text{coherent: } p_T \leq 0.1\,\text{GeV}/c, \qquad \text{incoherent: } 0.3 < p_T < 0.9\,\text{GeV}/c. \tag{13}$$

The incoherent range is a nominal choice; the final range is discussed with the mass spectrum (Section 5.2).

Figure 9 shows the pair p_T after the preselection. The ULS spectrum has a pronounced coherent peak below $0.1\,\text{GeV}/c$ that is absent in the LS pairs, and the two regions are well separated by the p_T cut alone. This separation depends only on the nuclear form factor and not on the produced vector meson, so it holds even though the sample is still dominated by pions at this stage.

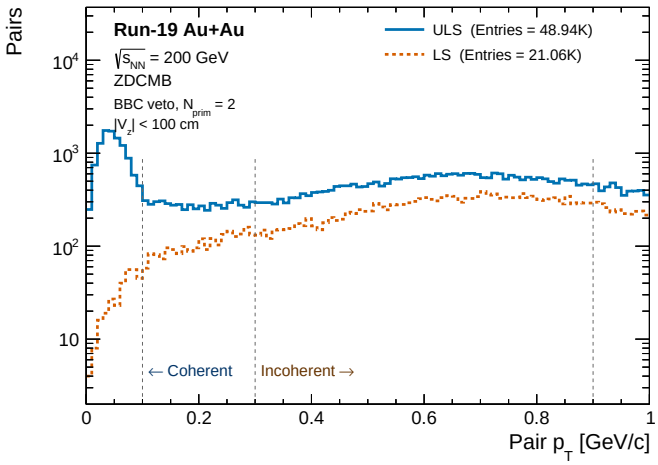


Figure 9: Pair p_T of ULS and LS pairs, same selection as Figure 5, with the coherent and incoherent regions of Equation 13.

4.3 Particle Identification

Figure 10 shows $\mathrm{d}E/\mathrm{d}x$ versus momentum for all tracks (ULS and LS) after the preselection. The curves give the expected $\mathrm{d}E/\mathrm{d}x$ for each hypothesis, taken as the mean $\mathrm{d}E/\mathrm{d}x$ of tracks with $|n\sigma_X| < 0.3$, i.e. the reference used in Equation 10. At low momentum the kaon band is well separated from the pion band, while the two merge above $p \approx 1\,\mathrm{GeV}/c$.

The sample is nevertheless dominated by pions. The preselection $\chi^2_{KK} < 100$ only limits each track to $|n\sigma_K| < 10$ (Figure 11), which at these momenta still accepts most of the pion band. This is also visible in the one-dimensional distributions: $n\sigma_\pi$ peaks sharply at zero, while $n\sigma_K$ is dominated by a broad pion contribution at negative values (Figure 12, Figure 13). A dedicated kaon selection is therefore required (Section 4.3.3, Section 4.3.4). Figure 11 also shows that the kaon band is centred at positive $n\sigma_K$ rather than at zero; this shift is measured with TOF-identified kaons in Section 4.3.1.

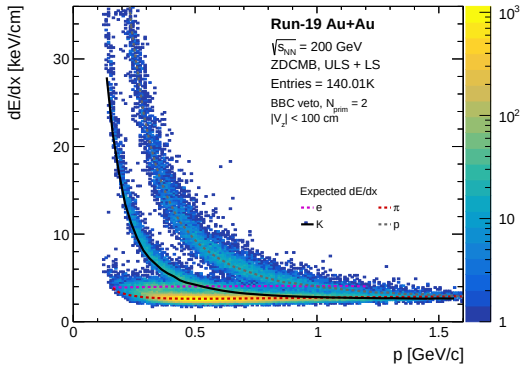


Figure 10: $\mathrm{d}E/\mathrm{d}x$ versus momentum for all tracks, same selection as Figure 5, with the expected $\mathrm{d}E/\mathrm{d}x$ for each hypothesis.

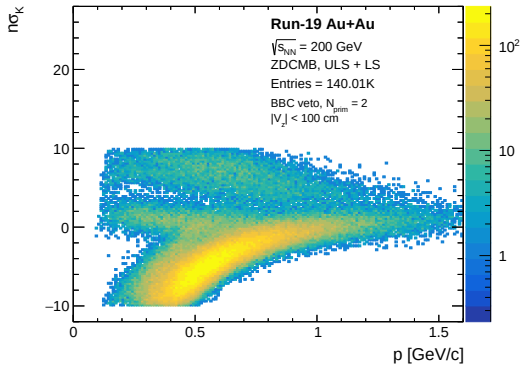


Figure 11: $n\sigma_K$ versus momentum for the same tracks.

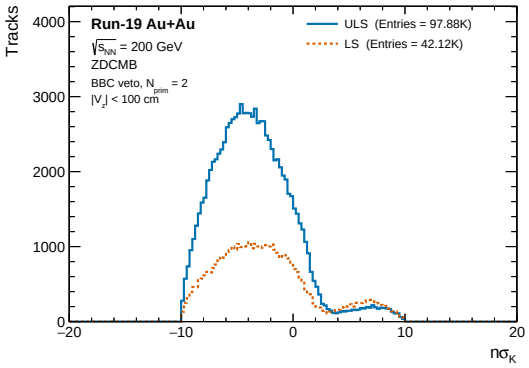


Figure 12: $n\sigma_K$ of tracks in ULS and LS pairs, same selection as Figure 5.

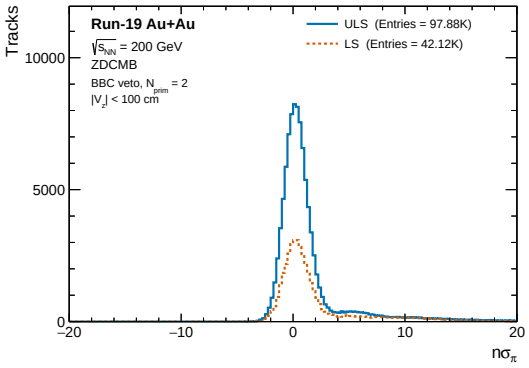


Figure 13: $n\sigma_\pi$ of the same tracks.

4.3.1 $n\sigma_K$ Calibration with TOF-Identified Kaons

The $n\sigma_K$ shift is measured with kaons identified independently of $\mathrm{d}E/\mathrm{d}x$ by the TOF. The $\phi \to K^+K^-$ UPC sample itself yields few TOF-identified kaons: the decay kaons are soft and mostly at large $|\eta|$, outside the TOF acceptance, and the sample is small. We therefore use kaons from peripheral hadronic Au+Au collisions in the same Run-19 data (trigger 700001, SL23e), which cover the full momentum range with high statistics. The sample is selected as follows:

- centrality 70–80% from `StRefMultCorr` (`getCentralityBin9() == 0`), $|V_z| < 50\,\mathrm{cm}$, $V_r < 2\,\mathrm{cm}$, bad runs removed;
- primary tracks of the highest-ranked vertex with $N_{\mathrm{hits}} \geq 20$ and $N_{\mathrm{hits}}/N_{\mathrm{poss}} > 0.52$;
- TOF-matched with $\beta < 1$ and $0.22 < m^2 < 0.27\,\mathrm{GeV}^2/c^4$, where $m^2 = p^2(1/\beta^2 - 1)$ (the stored sample extends to $0.28\,\mathrm{GeV}^2/c^4$);
- $|n\sigma_K| < 5$.

The TOF mass spectrum (Figure 14) shows well-separated pion, kaon, and proton peaks, and the m^2 window selects the kaon peak (Figure 15). The selected tracks form a single band in $n\sigma_K$ over the full momentum range (Figure 16), with only small pion and proton contamination at low momentum, confirming that the sample consists of genuine kaons.

The tracks are divided into momentum bins of $0.1\,\mathrm{GeV}/c$ between 0.1 and $2.5\,\mathrm{GeV}/c$, and the $n\sigma_K$ distribution in each bin is fitted with a Gaussian in $[-5, 5]$. The fitted mean $\langle n\sigma_K \rangle(p)$ (Figure 17) is positive at all momenta and largest at low momentum, consistent with the shift seen in Figure 11. We correct for it with

$$n\sigma_K^{\mathrm{corr}} = n\sigma_K - \langle n\sigma_K \rangle(p), \tag{14}$$

and $n\sigma_K^{\mathrm{corr}}$ is used in the kaon selection below.

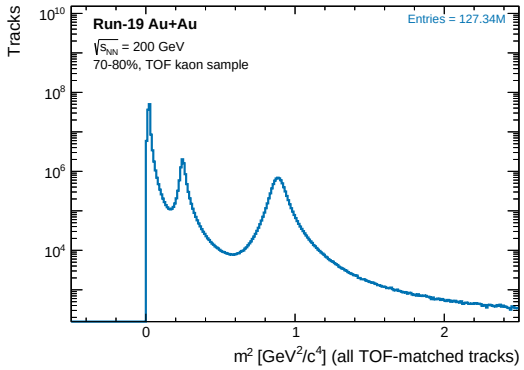


Figure 14: TOF m^2 of all TOF-matched tracks in the 70–80% sample.

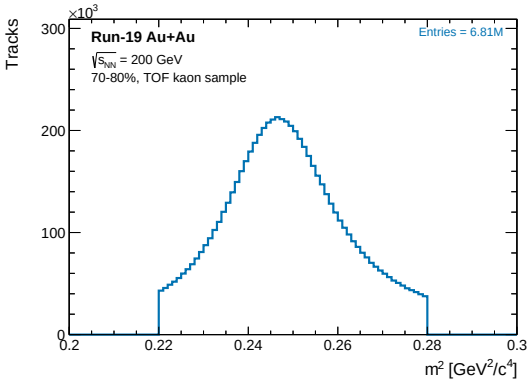


Figure 15: m^2 of the stored kaon candidates ($0.22 < m^2 < 0.28\,\mathrm{GeV}^2/c^4$).

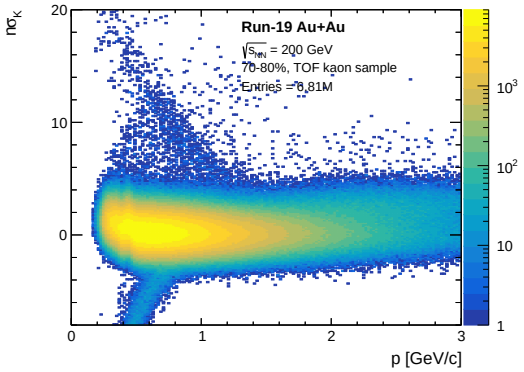


Figure 16: $n\sigma_K$ versus momentum for the stored kaon candidates, before the $|n\sigma_K| < 5$ requirement.

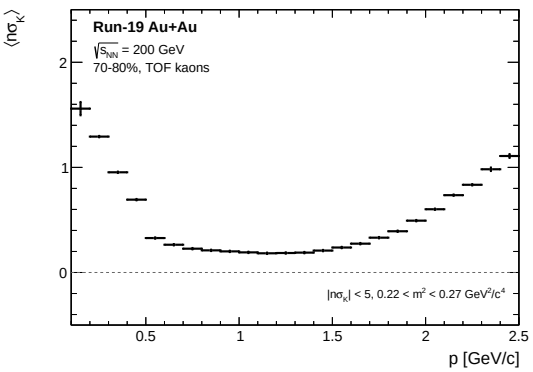


Figure 17: Mean $n\sigma_K$ of TOF-identified kaons versus momentum, from Gaussian fits.

The Gaussian fit in each momentum bin up to $p = 1.6\,\mathrm{GeV}/c$ is shown in Figure 18. The correction is applied in this range, which contains essentially all tracks of the ϕ sample; at higher momentum the kaon and pion bands merge and the fits become unreliable.

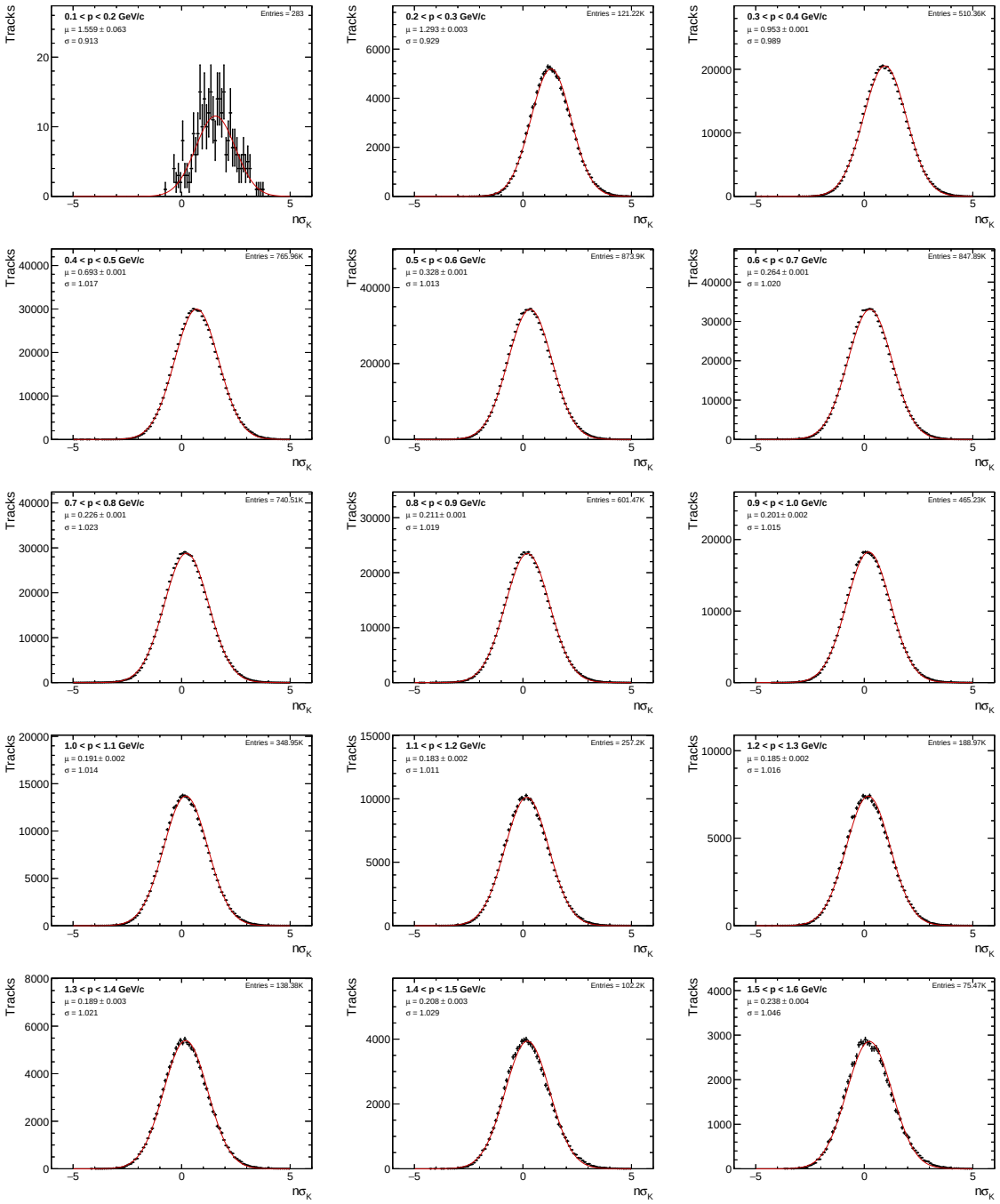


Figure 18: $n\sigma_K$ of TOF-identified kaons in each momentum bin with the Gaussian fit (red) used for Figure 17.

4.3.2 Cross-Check with Kaons from $\phi \to K^+K^-$

The $n\sigma_K$ shift is cross-checked in situ with kaons from ϕ decays in the UPC sample, selected as follows:

- BBC veto, two primary tracks in the event, $|V_z| \leq 100\,\text{cm}$, $V_r < 5\,\text{cm}$, pair rapidity $|y| < 0.5$, and $|n\sigma_K| \leq 5$ for both tracks;
- tag and probe: a track (probe) is used when the other track (tag) satisfies the pion rejection $|n\sigma_\pi| \geq 3$, so that no pion rejection is applied to the track whose $n\sigma_K$ is measured.

The non-ϕ background in the window $1.00 < M_{KK} < 1.04\,\text{GeV}/c^2$ is described by probes from the sideband $1.06 < M_{KK} < 1.10\,\text{GeV}/c^2$. The sideband is scaled to the background in the window obtained from a fit of the probe mass spectrum with a Voigtian (PDG width) plus a linear background (Figure 19). In each momentum bin, the $n\sigma_K$ distribution of the window probes is fitted with a binned Poisson likelihood using a Gaussian for the kaons plus the scaled sideband distribution as a fixed background template.

The resulting $\langle n\sigma_K \rangle(p)$ agrees with the TOF kaons within the statistical uncertainties (Figure 20). Since the TOF sample is far more precise and covers the full momentum range, the TOF result (Section 4.3.1) is used as the nominal correction.

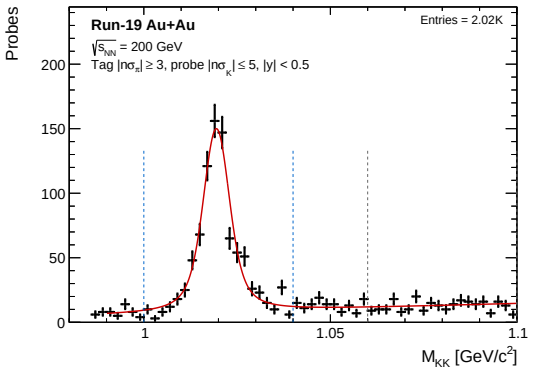


Figure 19: M_{KK} of tag-and-probe pairs with the fit; dashed lines mark the ϕ window and the sideband.

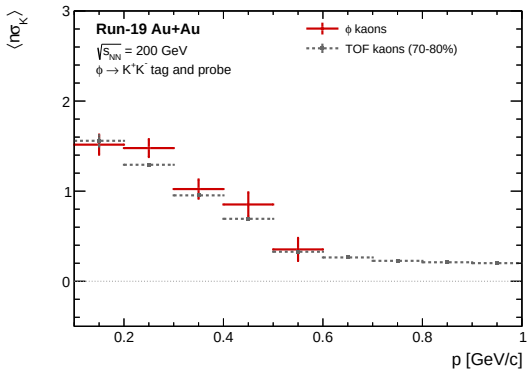


Figure 20: Mean $n\sigma_K$ of kaons from ϕ decays compared with the TOF-identified kaons.

4.3.3 Coherent Region (low pT)

4.3.4 Incoherent Region (high pT)

4.4 Selection Summary

5 Signal Extraction

5.1 Coherent Signal

5.1.1 Invariant-Mass Fit

5.1.2 Incoherent Contamination

5.2 Incoherent Signal

5.2.1 Invariant-Mass Fit

5.2.2 Background Estimation

5.3 Yield Summary

6 Efficiencies

6.1 Vertex Selection Efficiency

The vertex selection efficiency is the fraction of well-reconstructed vertices from the luminous region that lie inside the fiducial region $|V_z| < 100$ cm. It is measured with two samples: all UPC-like events (Section 6.1.1), whose V_z distribution has a broad background plateau, and a clean sample of coherent di-pion events (Section 6.1.2). Both give an efficiency consistent with unity, and $\varepsilon_{V_z} = 1.00$ is used as the nominal value (Section 6.1.3).

Both samples start from the following events, referred to as UPC-like events:

- `zdcmb` triggered, excluding bad runs.
- `bbcEast` $= 0$ and `bbcWest` $= 0$.
- Exactly two primary tracks matched to the same vertex.

6.1.1 Using All UPC-like Events

For all UPC-like events, there is no kinematic or particle-identification restriction on the reconstructed vertices, so secondary interactions are not rejected. Figure 21 and Figure 22 show signatures of interactions at $V_r \approx 3.8$ cm, which correspond to the beryllium beam pipe radius.

A second, sharp structure appears at $V_r = 10$ cm and extends up to $V_r = 10\sqrt{2} \approx 14.1$ cm. In Figure 22 these vertices lie on the edges of the square $|V_x| = 10$ cm or $|V_y| = 10$ cm, which is consistent with vertices pushed to the boundary of the transverse region allowed in the vertex reconstruction rather than interactions at a physical radius. These vertices carry no information on the luminous region and are not considered further.

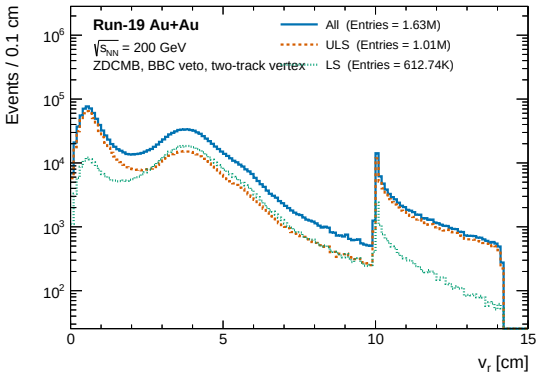


Figure 21: V_r distribution for all UPC-like events.

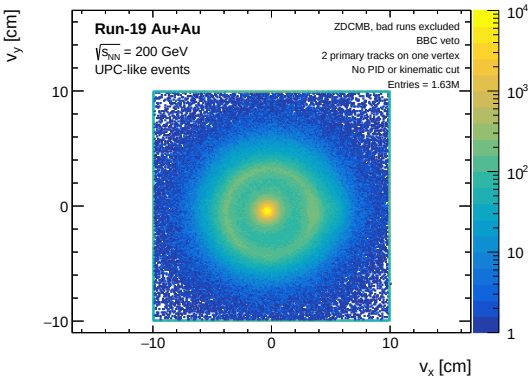


Figure 22: V_x versus V_y distribution for all UPC-like events.

To suppress secondary interactions for the efficiency study only, the following additional vertex selection criteria are implemented:

- $V_r < 3$ cm.
- Unlike-sign (ULS) pairs only.

Figure 23 shows that the additional vertex quality cuts remove this background. Figure 24 and Figure 25 show the V_z distribution before and after the vertex quality cuts.

The V_z distribution after the vertex quality cuts is fitted with the sum of two Gaussians,

$$f(V_z) = A_1 \exp\left(-\frac{(V_z-\mu_1)^2}{2\sigma_1^2}\right) + A_2 \exp\left(-\frac{(V_z-\mu_2)^2}{2\sigma_2^2}\right), \tag{15}$$

in 5 cm bins over $|V_z| < 190$ cm, which stays inside the ± 200 cm edge of the vertex reconstruction. The narrow Gaussian (σ_1) describes the luminous region, while the wide Gaussian (σ_2) absorbs the broad plateau seen in Figure 25. The fraction of a Gaussian inside the fiducial region is

$$F(\mu,\sigma) = \frac{1}{2}\left[\mathrm{erf}\left(\frac{100-\mu}{\sqrt{2}\sigma}\right) - \mathrm{erf}\left(\frac{-100-\mu}{\sqrt{2}\sigma}\right)\right], \tag{16}$$

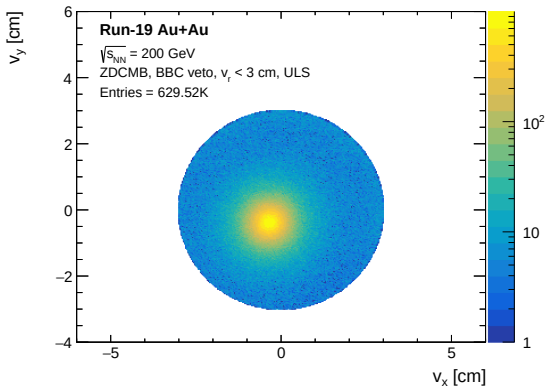


Figure 23: V_x versus V_y distribution for unlike-sign pairs in UPC-like events with $V_r < 3$ cm.

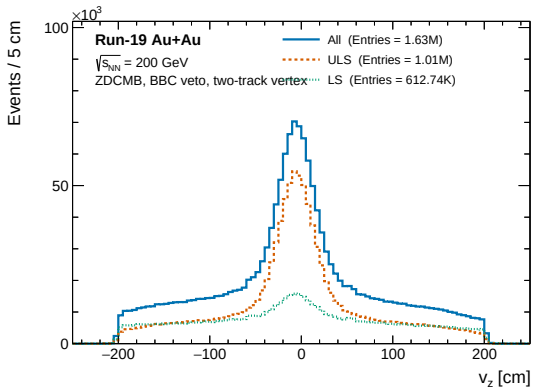


Figure 24: V_z distribution for all UPC-like events.

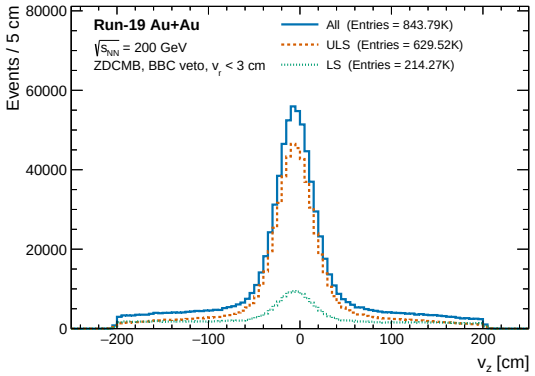


Figure 25: V_z distribution for UPC-like events with $V_r < 3$ cm.

from which three estimates of the fraction inside $|V_z| < 100$ cm are formed:

- $f_{\text{narrow}} = F(\mu_1, \sigma_1)$, from the narrow Gaussian only;
- $f_{\text{both}} = (1 - w_2)F(\mu_1, \sigma_1) + w_2 F(\mu_2, \sigma_2)$, from both Gaussians, where $w_2 = A_2\sigma_2/(A_1\sigma_1 + A_2\sigma_2)$ is the area fraction of the wide Gaussian;
- f_{count}, the fraction of entries with $|V_z| < 100$ cm, counted directly from the histogram.

The fits to the unlike-sign pairs (Figure 26) and, as a cross-check, to the unlike-sign minus like-sign (ULS − LS) distribution (Figure 27) are summarized in Table 3.

Table 3: Double-Gaussian fit results for UPC-like events with $V_r < 3$ cm. All fractions refer to $|V_z| < 100$ cm.

Sample	σ_1 [cm]	σ_2 [cm]	w_2	f_{narrow}	f_{both}	f_{count}
ULS	19.9	124.3	0.372	1.0000	0.843	0.879
ULS − LS	19.0	64.1	0.242	1.0000	0.971	0.975

The narrow Gaussian is stable between the two samples ($\sigma_1 \approx 19$–20 cm, $\mu_1 \approx -5.4$ cm) and lies entirely inside the fiducial region, $f_{\text{narrow}} = 1.0000$. The wide Gaussian, in contrast, changes substantially when the like-sign pairs are subtracted (σ_2 from 124 to 64 cm, w_2 from 0.37 to 0.24), showing that it is driven by the residual background plateau rather than by the luminous region. The lower values of f_{both} and f_{count} therefore reflect background that extends beyond $|V_z| = 100$ cm and are not used as the efficiency. The vertex selection efficiency from this method is taken from the narrow Gaussian, $\varepsilon_{V_z} \approx 1.000$. Both fits have large χ^2/ndf (964/70 and 340/70), as two Gaussians do not fully describe the peak-plus-plateau shape; this does not affect f_{narrow}, which is set by $\sigma_1 \approx 20$ cm relative to the 100 cm cut.

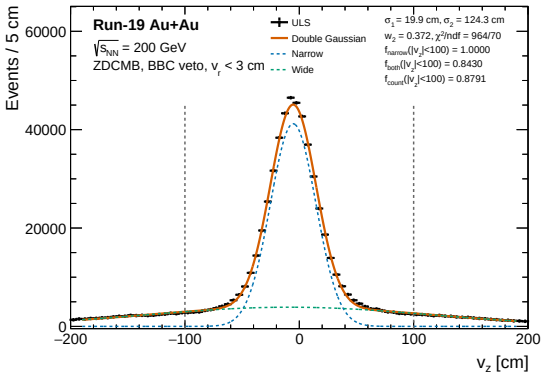


Figure 26: Double-Gaussian fit to the V_z distribution of unlike-sign pairs in UPC-like events with $V_r < 3$ cm. Dashed vertical lines mark $|V_z| = 100$ cm.

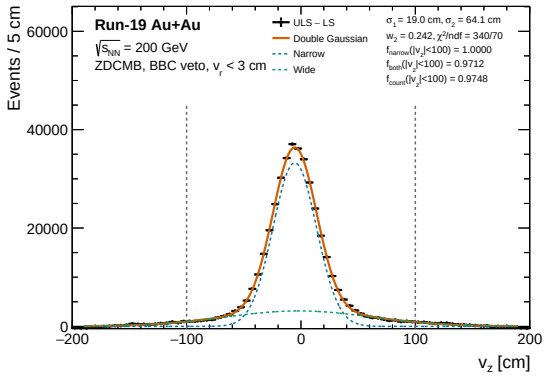


Figure 27: Double-Gaussian fit to the unlike-sign minus like-sign V_z distribution in UPC-like events with $V_r < 3$ cm. Dashed vertical lines mark $|V_z| = 100$ cm.

6.1.2 Using Di-pion Events

To avoid the background plateau of the UPC-like events, the luminous region is also measured with a clean sample of coherently photoproduced $\pi^+\pi^-$ pairs. In addition to the UPC-like event definition, the following selection is applied:

- `TofMult` ≤ 6 and at least one track matched to the TOF.
- Both tracks satisfy $|n\sigma_\pi| < 3$.
- Pair transverse momentum $p_T^{\pi\pi} < 0.1$ GeV/c (coherent region).
- $0.60 < M_{\pi\pi} < 0.95$ GeV/c^2 around the ρ^0 pole, for the vertex distributions only.

The vertex coordinates are those of the vertex the two pions are matched to.

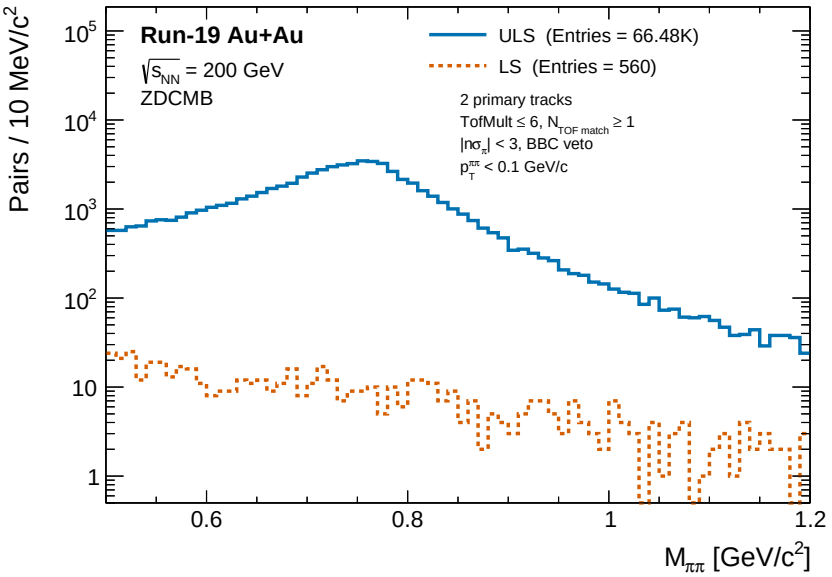
6.1.2.1 Di-pion Mass Spectrum

The mass spectrum in this section is not corrected for acceptance or efficiency and is not used to extract a cross section. It only serves to check that the selected di-pion sample is not background dominated and that its shape agrees with the published coherent ρ^0 result.

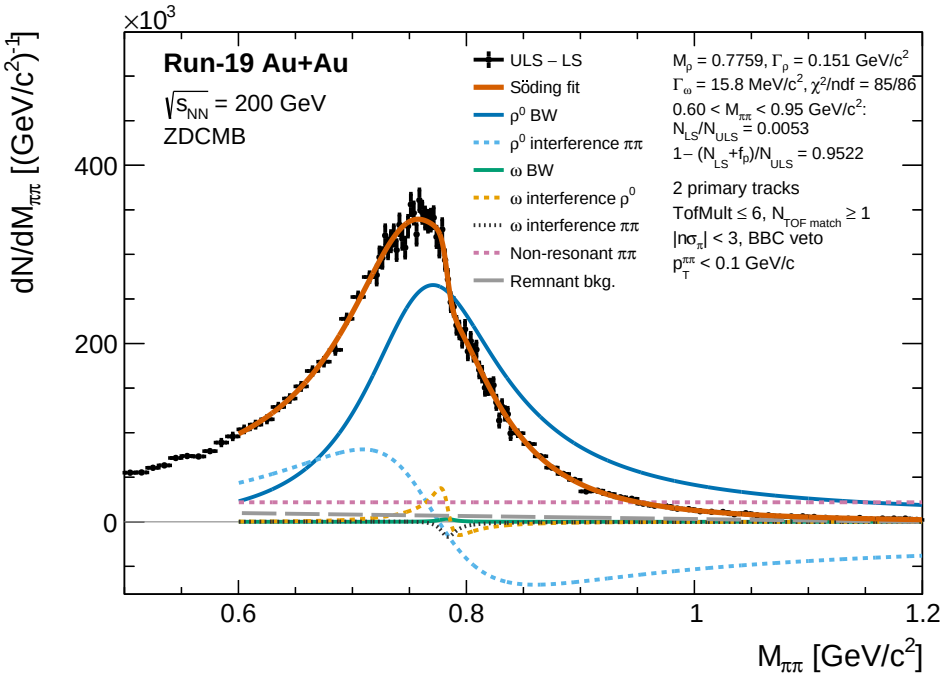
Figure 28 compares the unlike-sign and like-sign invariant-mass spectra of the selected pairs. The like-sign pairs are about two orders of magnitude below the unlike-sign pairs over the whole mass range. The unlike-sign minus like-sign spectrum is fitted over $0.6 < M_{\pi\pi} < 1.2$ GeV/c^2 with the modified Söding parametrization used by STAR for coherent ρ^0 photoproduction (Phys. Rev. C 96 (2017) 054904),

$$\frac{dN}{dM_{\pi\pi}} \propto \left| A_\rho \frac{\sqrt{M_{\pi\pi} M_\rho \Gamma_\rho}}{M_{\pi\pi}^2 - M_\rho^2 + i M_\rho \Gamma_\rho} + B_{\pi\pi} + C_\omega e^{i\varphi_\omega} \frac{\sqrt{M_{\pi\pi} M_\omega \Gamma_{\omega\to\pi\pi}}}{M_{\pi\pi}^2 - M_\omega^2 + i M_\omega \Gamma_\omega} \right|^2 + f_p, \quad (17)$$

with mass-dependent widths as in that reference and a linear remnant background f_p. The spectrum uses 2.5 MeV/c^2 bins in $0.72 < M_{\pi\pi} < 0.84$ GeV/c^2, where the ρ^0–ω interference is located, and 10 MeV/c^2 bins elsewhere. The fit and its components are shown in Figure 29 and the parameters are compared with the published values in Table 4.



408 Figure 28: Invariant-mass spectra of unlike-sign and like-sign pion pairs passing the di-pion
409 selection, without vertex or mass-window requirements.



410 Figure 29: Modified Söding fit to the unlike-sign minus like-sign $\pi^+\pi^-$ spectrum, with the
411 ρ^0, ω, interference, non-resonant and remnant-background components.

Table 4: Modified Söding fit to the coherent unlike-sign minus like-sign $\pi^+\pi^-$ spectrum, compared with the published fit. The published ratios are computed from the quoted amplitudes.

Parameter	This analysis	STAR, PRC 96, 054904
M_ρ [GeV/c^2]	0.7759	0.7762 ± 0.0006
Γ_ρ [GeV/c^2]	0.151	0.156 ± 0.001
$B_{\pi\pi}/A_\rho$ [(GeV/c^2)$^{-1/2}$]	-0.742	-0.787
C_ω/A_ρ	0.295	0.358
M_ω [GeV/c^2]	0.7829	0.7824 ± 0.0008
Γ_ω [MeV/c^2]	15.8	17 ± 2
φ_ω [rad]	1.33	1.46 ± 0.11
χ^2/ndf	85.5/86	–

The ρ^0 mass and width and the ω parameters are close to the published values. As the spectrum is uncorrected and no systematic uncertainties are evaluated, this comparison is qualitative: it shows that the lineshape is reproduced. As in the published fit, no mass resolution is included in the model, so the fitted Γ_ω absorbs the detector resolution and is about twice the natural width. The ratio $B_{\pi\pi}/A_\rho = -0.742$ is also close to the published -0.787, showing that the non-resonant $\pi^+\pi^-$ component is described by $B_{\pi\pi}$ rather than absorbed into the remnant background f_p. Inside the mass window $0.60 < M_{\pi\pi} < 0.95$ GeV/c^2 there are 56788 unlike-sign and 301 like-sign pairs ($N_{\rm LS}/N_{\rm ULS} = 0.0053$), and the remnant background f_p amounts to 2415 pairs, giving a $\pi^+\pi^-$ purity of $1 - (N_{\rm LS} + f_p)/N_{\rm ULS} = 0.952$. The selected sample is therefore dominated by coherent $\pi^+\pi^-$ production.

6.1.2.2 Vertex Distributions and Efficiency

Figure 30 and Figure 31 show the transverse position of the di-pion vertices. Unlike the UPC-like events, there is no ring at $V_r \approx 3.8$ cm: the vertices are concentrated around the beam spot, with only a small population at the $V_r = 10$ cm reconstruction boundary. The V_z distribution (Figure 32) shows no plateau, and the like-sign contribution is negligible.

The V_z distributions are fitted with the same double-Gaussian model as for the UPC-like events, for unlike-sign pairs and for unlike-sign minus like-sign pairs, with and without the $V_r < 3$ cm requirement (Table 5). The fits for unlike-sign pairs without a V_r requirement and for unlike-sign minus like-sign pairs with $V_r < 3$ cm are shown in Figure 33 and Figure 34.

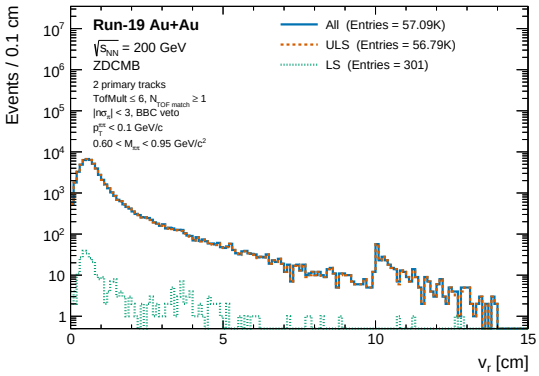


Figure 30: V_r distribution of coherent $\pi^+\pi^-$ pairs.

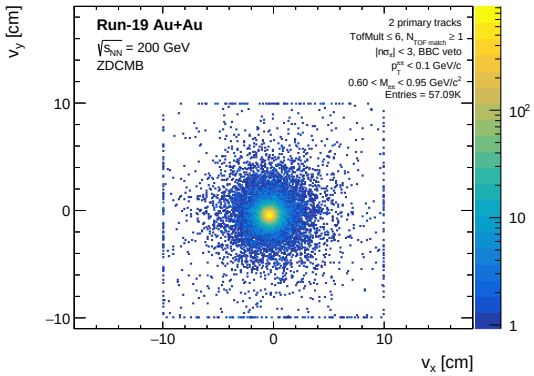


Figure 31: V_x versus V_y distribution of coherent $\pi^+\pi^-$ pairs.

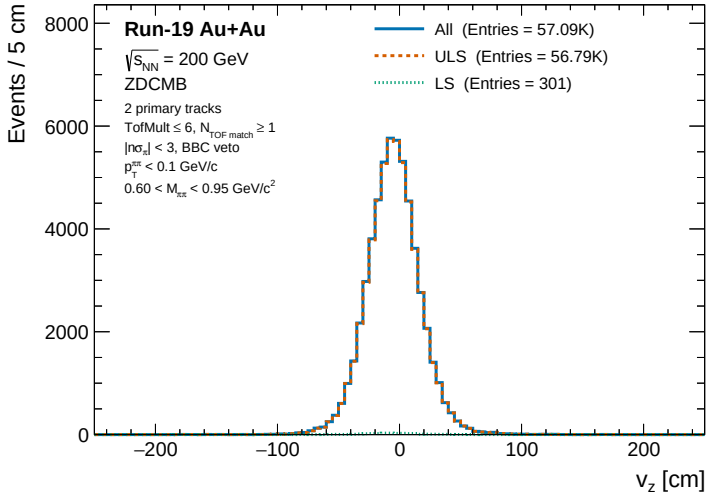


Figure 32: V_z distribution of coherent $\pi^+\pi^-$ pairs.

Table 5: Double-Gaussian fit results for coherent $\pi^+\pi^-$ pairs. In all cases $f_{\mathrm{narrow}} = 1.0000$. All fractions refer to $|V_z| < 100$ cm.

Sample	V_r cut	σ_1 [cm]	σ_2 [cm]	w_2	χ^2/ndf	f_{both}	f_{count}
ULS	none	18.6	33.3	0.155	81/57	0.9996	0.9990
ULS − LS	none	18.5	32.4	0.169	79/60	0.9996	0.9993
ULS	< 3 cm	18.5	32.2	0.168	56/48	0.9997	0.9995
ULS − LS	< 3 cm	18.5	32.0	0.172	56/48	0.9997	0.9995

In contrast to the UPC-like events, the wide Gaussian is the same in all four cases ($\sigma_2 \approx$ 32–33 cm, $w_2 \approx 0.16$–0.17): it does not depend on the like-sign subtraction or on the V_r requirement and is therefore part of the luminous region rather than background. Both Gaussians are thus included, and the fits describe the data well ($\chi^2/\mathrm{ndf} \approx 1.2$–$1.4$). The

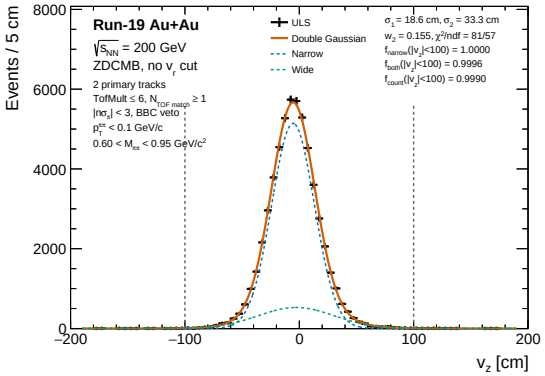


Figure 33: Double-Gaussian fit to the V_z distribution of coherent unlike-sign $\pi^+\pi^-$ pairs without a V_r requirement. Dashed vertical lines mark $|V_z| = 100$ cm.

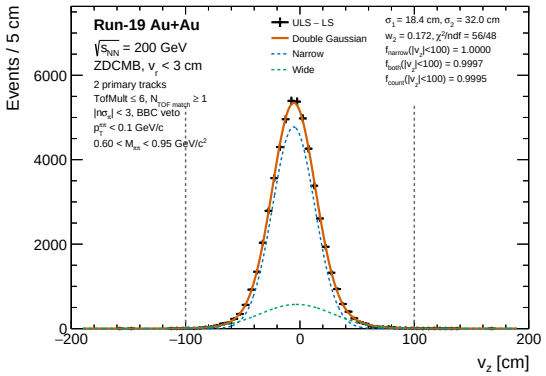


Figure 34: Double-Gaussian fit to the unlike-sign minus like-sign V_z distribution of coherent $\pi^+\pi^-$ pairs with $V_r < 3$ cm. Dashed vertical lines mark $|V_z| = 100$ cm.

wide Gaussian lies almost entirely inside the fiducial region, so f_{both}, f_{narrow} and f_{count} agree within 0.001. The vertex selection efficiency from the di-pion sample is $\varepsilon_{V_z} = 0.9996$, with a spread of 0.0006 among the variations in Table 5, consistent with the narrow-Gaussian result from the UPC-like events.

6.1.3 Nominal Vertex Selection Efficiency

Both methods give a vertex selection efficiency consistent with unity: the narrow Gaussian of the UPC-like events gives $f_{\text{narrow}} = 1.000$, and the coherent $\pi^+\pi^-$ sample gives $\varepsilon_{V_z} = 0.9996$, with all variations within 0.001 of unity. We therefore use $\varepsilon_{V_z} = 1.00$ as the nominal value.

6.2 BBC Veto Selection Efficiency

The BBC veto (Equation 9) rejects events with activity in the small BBC tiles. Besides hadronic background, it also removes genuine UPC events that are accompanied by additional BBC activity, either from the same collision (e.g. electromagnetic dissociation of the nuclei) or from pile-up. The corresponding efficiency is measured with coherently photoproduced $\pi^+\pi^-$ pairs, as the ratio of the ρ^0 yields with and without the veto.

The di-pion selection of the vertex study is used, with two changes: the `TofMult` ≤ 6 requirement is removed, since TOF activity is correlated with BBC activity and such a cut would bias the veto efficiency, and no V_z requirement is applied. The selection is therefore exactly two primary tracks on one vertex, at least one track matched to the TOF, $|n\sigma_\pi| < 3$ for both tracks and $p_T^{\pi\pi} < 0.1$ GeV/c as a proxy for coherent selection.

The efficiency can only be measured on a sample that remains signal dominated without the BBC veto. This is ensured by the selection itself. Exactly two primary tracks identified as pions are required, and $\pi^+\pi^-$ photoproduction (dominantly ρ^0) is the most abundant exclusive final state in UPC. In addition, $p_T^{\pi\pi} < 0.1$ GeV/c restricts the sample to the coherent peak, a narrow window in pair transverse momentum where hadronic and

incoherent contributions are small. As a result, even without the BBC veto the like-sign pairs amount to only $N_{\mathrm{LS}}/N_{\mathrm{ULS}} = 1.0\%$ in $0.6 < M_{\pi\pi} < 1.2$ GeV/c^2, compared with 0.64% with the veto.

The unlike-sign minus like-sign mass spectra of all selected events, of the events passing the BBC veto, and of the events rejected by it are fitted over $0.6 < M_{\pi\pi} < 1.2$ GeV/c^2 with the modified Söding parametrization (Equation 17), using the same binning and fit procedure as in the vertex study (Figure 35, Figure 36, Figure 37). All three fits describe the data well and their shape parameters agree within uncertainties (Table 6), so the veto changes the yield but not the shape of the spectrum.

None of the samples is background dominated. The remnant background f_p amounts to 5.8% of the unlike-sign minus like-sign pairs for all events and 4.8% for events passing the veto. The slightly larger background without the veto is expected, since the veto preferentially removes hadronic background: only about half of the like-sign pairs (750 to 379) survive the veto, compared with 82% of the signal. Two yields are extracted from each fit: the integral of the fit without the remnant background f_p over the fit range, and the integral of the ρ^0 Breit-Wigner from $2m_\pi$ to $M_\rho + 5\Gamma_\rho$.

The fit to the rejected events (Figure 37) shows that these are predominantly genuine UPC-like di-pions rather than hadronic background. Their spectrum shows the same ρ^0 lineshape, with M_ρ, Γ_ρ and $B_{\pi\pi}/A_\rho$ consistent with those of the accepted events and the ρ^0–ω interference visible at $M_{\pi\pi} \approx 0.78$ GeV/c^2. The background in this sample is larger but still small: $N_{\mathrm{LS}}/N_{\mathrm{ULS}} = 2.7\%$ and f_p amounts to 10% of the unlike-sign minus like-sign pairs. The BBC veto therefore removes genuine UPC signal.

The efficiency is computed from the two disjoint samples as $\varepsilon_{\mathrm{BBC}} = Y_{\mathrm{pass}}/\left(Y_{\mathrm{pass}} + Y_{\mathrm{rejected}}\right)$. Since the samples do not overlap, their fit uncertainties are independent and are propagated directly; a binomial uncertainty on $Y_{\mathrm{pass}}/Y_{\mathrm{all}}$ would neglect the uncertainty of the fit decomposition, which dominates the yield uncertainties. The efficiency is

- $\varepsilon_{\mathrm{BBC}} = 0.826 \pm 0.004$ from the yield without f_p,

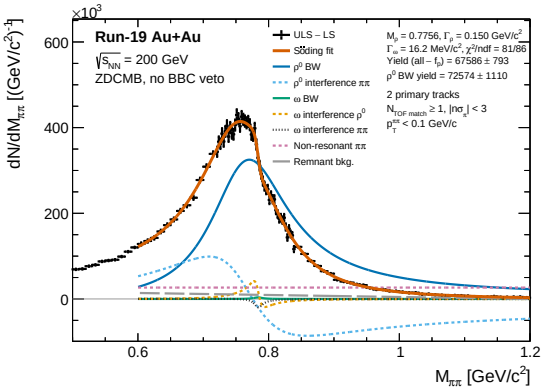


Figure 35: Modified Söding fit to the unlike-sign minus like-sign $\pi^+\pi^-$ spectrum without the BBC veto, with all fit components.

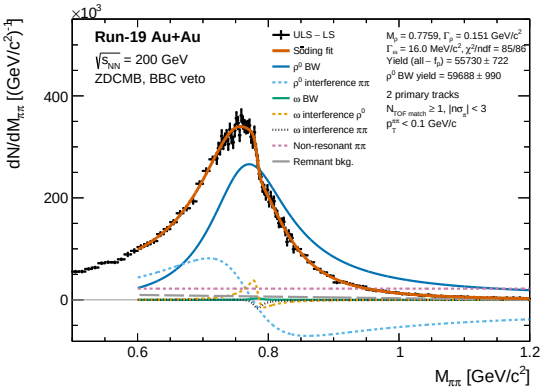


Figure 36: Modified Söding fit to the unlike-sign minus like-sign $\pi^+\pi^-$ spectrum with the BBC veto, with all fit components.

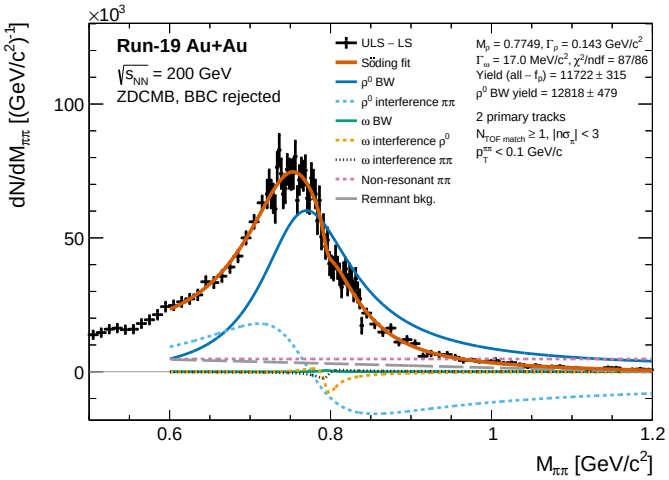


Figure 37: Modified Söding fit to the unlike-sign minus like-sign $\pi^+\pi^-$ spectrum of the events rejected by the BBC veto, with all fit components.

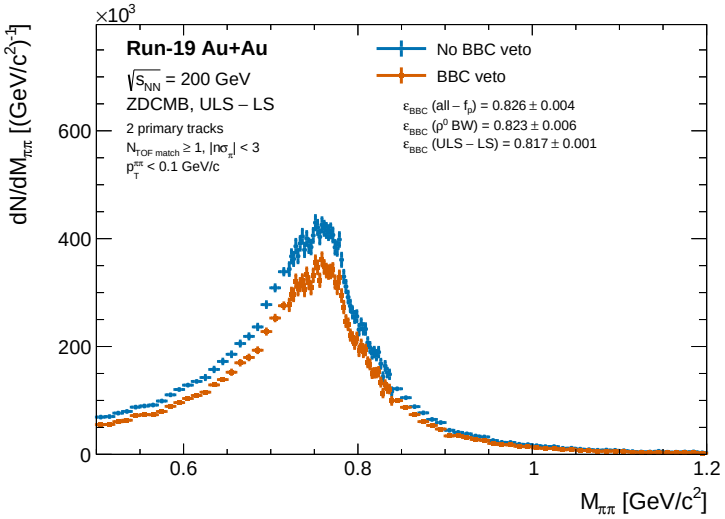
- $\varepsilon_{\mathrm{BBC}} = 0.823 \pm 0.006$ from the ρ^0 Breit-Wigner yield,
- $\varepsilon_{\mathrm{BBC}} = 0.817 \pm 0.002$ from a direct unlike-sign minus like-sign count over the fit range.

The sum of the passing and rejected yields reproduces the yield of all events to 0.2%, and the ratio $Y_{\mathrm{pass}}/Y_{\mathrm{all}}$ gives consistent values (0.825 and 0.822 for the two fit-based yields). The direct count is lower because it includes the remnant background, which is preferentially removed by the veto. The spectra with and without the veto are compared in Figure 38.

To check whether the efficiency depends on the kinematics, it is computed in bins of $M_{\pi\pi}$ and of the pair rapidity $y_{\pi\pi}$ within the ρ^0 window $0.60 < M_{\pi\pi} < 0.95$ GeV/c^2, from unlike-sign minus like-sign counts with binomial uncertainties (Figure 39, Figure 40). The efficiency does not vary with the kinematics: compared with the efficiency of the whole window, the bins give $\chi^2 = 8.4$ for 7 bins in $M_{\pi\pi}$ (0.809–0.823) and $\chi^2 = 2.4$ for 8 bins

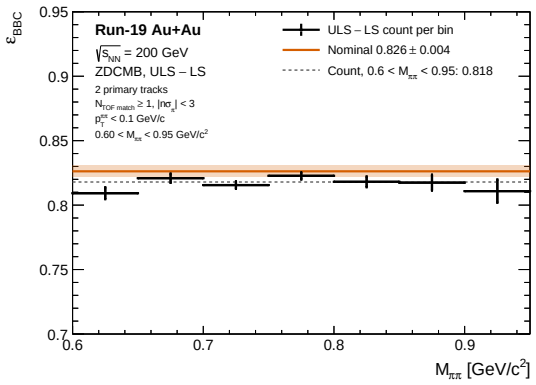
Table 6: Modified Söding fits to the coherent $\pi^+\pi^-$ spectrum for all selected events, for events passing the BBC veto, and for events rejected by it. Uncertainties are statistical, from the fits; the uncertainties of $B_{\pi\pi}/A_\rho$ and C_ω/A_ρ include the correlation with A_ρ.

	All events	Pass BBC veto	Rejected by BBC veto
M_ρ [GeV/c^2]	0.7756 ± 0.0012	0.7759 ± 0.0013	0.7749 ± 0.0025
Γ_ρ [GeV/c^2]	0.150 ± 0.003	0.151 ± 0.004	0.143 ± 0.006
$B_{\pi\pi}/A_\rho$ [(GeV/c^2)$^{-1/2}$]	-0.740 ± 0.017	-0.742 ± 0.018	-0.744 ± 0.040
C_ω/A_ρ	0.283 ± 0.049	0.295 ± 0.056	0.249 ± 0.103
Γ_ω [MeV/c^2]	16.2 ± 4.1	16.0 ± 4.5	17.0 ± 8.2
χ^2/ndf	81/86	85/86	87/86
Yield (all $- f_p$)	67586 ± 793	55730 ± 722	11722 ± 315
ρ^0 BW yield	72574 ± 1110	59688 ± 990	12818 ± 479



542 Figure 38: Unlike-sign minus like-sign $\pi^+\pi^-$ spectra with and without the BBC veto, with
543 the resulting efficiencies.

544 in $y_{\pi\pi}$ (0.814–0.821). The binned counts lie about 0.008 below the nominal value because,
545 like the integrated count, they include the remnant background that the veto removes
546 preferentially. The pair transverse momentum is not used for this check. The selection is
547 restricted to the coherent region, where the sample is clean; towards larger $p_T^{\pi\pi}$ the hadronic
548 background grows, so a dependence on $p_T^{\pi\pi}$ would reflect the changing background fraction
549 rather than the efficiency for the signal.



550 Figure 39: BBC veto efficiency versus $M_{\pi\pi}$
551 from unlike-sign minus like-sign counts. The
552 solid line and band show the nominal value
553 and its uncertainty, the dashed line the effi-
554 ciency from the integrated count over the
555 window.

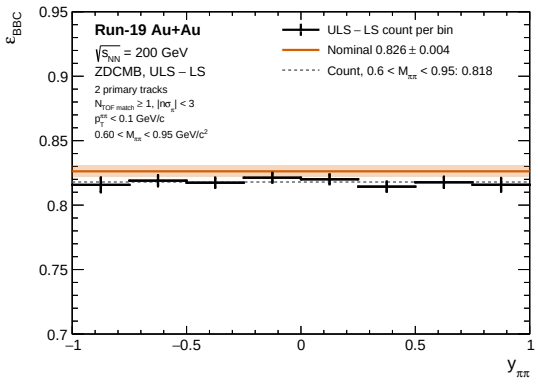


Figure 40: BBC veto efficiency versus the pair rapidity $y_{\pi\pi}$, for $0.60 < M_{\pi\pi} < 0.95$ GeV/c^2. Lines as in Figure 39.

6.2.1 Nominal BBC Veto Efficiency

We use $\varepsilon_{\mathrm{BBC}} = 0.826 \pm 0.004$ as the nominal value, from the yield without f_p, where the uncertainty is statistical, from the fits. The ρ^0 Breit-Wigner yield gives a consistent result.